OOP Basics

Task 1: [Homework]

Consider the following class:

```
public class Human{
    public int age;
    public double height;
}
```

Show the output of the following sequence of statements:

	Output
`Human h1 = new Human();`	
`Human h2 = new Human();`	
`h1.age = 21;`	
`h1.height = 5.5;`	
`System.out.println(h1.age);`	
`System.out.println(h1.height);`	
`h2.height = h1.height - 3;`	
`System.out.println(h2.height);`	
`h2.age = h1.age++;`	
`System.out.println(h1.age);`	
`h2 = h1;`	
`System.out.println(h2.age);`	
`System.out.println(h2.height);`	
`h2.age++;`	
`h2.height++;`	
`System.out.println(h1.age);`	
`System.out.println(h1.height);`	
`h1.age = ++h2.age;`	
`System.out.println(h2.age);`	
`System.out.println(h2.height);`	

Task 2:

Design the necessary class to generate the correct output from the driver code provided below:

Driver Code	Output

`public class CourseTester{`
`  public static void main(String args []){`
`  Course c1 = new Course();`
`  System.out.println("Course Name: "+c1.courseName);`
`  System.out.println("Course Code: "+c1.courseCode);`
`  System.out.println("Credit: "+c1.credit);`
`  }`
`}`
	`Course Name: null`
	`Course Code: null`
	`Credit: 0`

Task 3 :

Design the class to generate the correct output from the driver code provided below:

Driver Code	Output

`public class CourseTester{`
`  public static void main(String args []){`
`  Course c1 = new Course();`
`  System.out.println("Course Name: "+c1.courseName);`
`  System.out.println("Course Code: "+c1.courseCode);`
`  System.out.println("Credit: "+c1.credit);`
`  }`
`}`
	`Course Name: Programming Language II`
	`Course Code: CSE111`
	`Credit: 3`

Task 4: [Homework]

Design the class to generate the correct output from the driver code provided below:

Driver Code	Output

`public class PlayerTester{`
`  public static void main(String args[]){`
`  Player player1 = new Player();`
`  player1.name = "Ronaldo";`
`  player1.jersy_number = 9;`
`  player1.position = "Striker";`
	Name of the Player: Ronaldo
	Jersey Number of player: 9
	Position of player: Striker
	==========
	Name of the player: Neuer

System.out.println("Name of the Player: "+ player1.name);
System.out.println("Jersey Number of player: "+ player1.jersy_number);
System.out.println("Position of player: "+ player1.position);
System.out.println("==========");
Player player2 = new Player();
player2.name = "Neuer";
player2.jersy_number = 1;
player2.position = "Goal Keeper";
System.out.println("Name of the player: "+ player2.name);
System.out.println("Jersey Number of player: "+ player2.jersy_number);
System.out.println("Position of player: "+ player2.position);
}
}
	Jersey Number of player: 1
	Position of player: Goal Keeper

Instance Method

Task 5:

Design the **Phone** class to generate the correct output from the driver code provided below:

Driver Code	Output

`public class PhoneTester{`
`  public static void main(String args[]){`
`  Phone p1 = new Phone();`
`  p1.showDetails();`
`  System.out.println("---------1---------");`
`  System.out.println(p1.increasePrice(2000));`
`  System.out.println("---------2---------");`
`  p1.showDetails();`
`  Phone p2 = new Phone();`
`  System.out.println("---------4---------");`
`  p2.showDetails();`
`  System.out.println("---------5---------");`
`  System.out.println(p2.increasePrice(5000));`
`  System.out.println("---------6---------");`
`  p2.changeBrandName("Oppo");`
`  System.out.println("---------7---------");`
`  p2.showDetails();`
`  }`
`}`
	`Brand Name: Nokia`
	`Color: Blue`
	`Price: 14000 Taka`
	`---------1---------`
	`The price is updated`
	`---------2---------`
	`Brand Name: Nokia`
	`Color: Blue`
	`Price: 16000 Taka`
	`---------4---------`
	`Brand Name: Nokia`
	`Color: Blue`
	`Price: 14000 Taka`
	`---------5---------`
	`The price is updated`
	`---------6---------`
	`Brand name is changed to Oppo`
	`---------7---------`
	`Brand Name: Oppo`
	`Color: Blue`
	`Price: 19000 Taka`

Task 6: **[Homework]**

Design the necessary class to generate the correct output from the driver code provided below:

Driver Code	Output

`public class VehicleTester{`
`  public static void main (String args[]){`
`  Vehicle v1 = new Vehicle();`
`  v1.x = 0;`
`  v1.y = 0;`
`  System.out.println(v1.details());`
`  System.out.println("============");`
`  v1.moveUp();`
	`0,0`
	`============`
	`0,1`
	`============`
	`-1,0`
	`============`
	`0,0`
	`============`

```
    System.out.println(v1.details());
    System.out.println("============");
    v1.moveLeft();
    v1.moveDown();
    System.out.println(v1.details());
    System.out.println("============");
    v1.moveRight();
    System.out.println(v1.details());
    System.out.println("============");
  }
}
```

Task 7:

Design the **Student** class to generate the correct output from the driver code provided below: **Hint: A student can add a maximum of 4 courses.**

Driver Code	Output

public class StudentTester{
public static void main (String args[]){
Student std1 = new Student();
std1.studentInfo();
System.out.println("1----------");
std1.name = "Bob";
std1.id = 123;
System.out.println(std1.addCourse("CSE110"));
System.out.println("2----------");
std1.studentInfo();
System.out.println("3----------");
Student std2 = new Student();
std2.name = "Max";
std2.id = 456;
System.out.println(std2.addCourse("CSE250"));
System.out.println(std2.addCourse("CSE370"));
std2.studentInfo();
System.out.println("4----------");
System.out.println(std2.addCourse("PHY111"));
System.out.println(std2.addCourse("MAT120"));
System.out.println("5----------");
std2.studentInfo();
System.out.println("6----------");
	Student Name: Default
	Student ID: 0
	Maximum Course Limit: 4
	Courses Taken: 0
	Courses:
	1----------
	CSE110 course is added
	2----------
	Student Name: Bob
	Student ID: 123
	Maximum Course Limit: 4
	Courses Taken: 1
	Courses: CSE110
	3----------
	CSE250 course is added
	CSE370 course is added
	Student Name: Max
	Student ID: 456
	Maximum Course Limit: 4
	Courses Taken: 2
	Courses: CSE250 CSE370
	4----------
	PHY111 course is added

System.out.println(std2.addCourse("CSE470"));
System.out.println("7----------");
std2.studentInfo();
 	
}
}
	MAT120 course is added
	5----------
	Student Name: Max
	Student ID: 456
	Maximum Course Limit: 4
	Courses Taken: 4
	Courses: CSE250 CSE370 PHY111 MAT120
	6----------
	The maximum course limit exceeded
	7----------
	Student Name: Max
	Student ID: 456
	Maximum Course Limit: 4
	Courses Taken: 4
	Courses: CSE250 CSE370 PHY111 MAT120

Task 8:

Write the design class with the required properties to produce the given output for the provided driver code.

Driver Code	Output

public class TaxiTester{
public static void main(String args[]){
Taxi t1 = new Taxi();
t1.createTaxi(4);
System.out.println("1----------");
t1.info();
System.out.println("2----------");
t1.addPassenger("Walker", 200);
t1.addPassenger("Matt", 200);
System.out.println("3----------");
t1.info();
System.out.println("4----------");
t1.addPassenger("Wilson",400);
System.out.println("5----------");
t1.info();
System.out.println("5----------");
t1.addPassenger("Henry",150);
System.out.println("6----------");
t1.info();
System.out.println("7----------");
t1.addPassenger("Marry", 200);
System.out.println("8----------");
	A taxi is created with a capacity of 4
	1----------
	Total Passenger: 0
	Maximum Capacity: 4
	Total Fare: 0 Taka
	Passenger List:
	2----------
	Dear Walker! Welcome to Taxi
	Dear Matt! Welcome to Taxi
	3----------
	Total Passenger: 2
	Maximum Capacity: 4
	Total Fare: 400 Taka
	Passenger List:
	Walker Matt
	4----------
	Dear Wilson! Welcome to Taxi
	5----------
	Total Passenger: 3
	Maximum Capacity: 4
	Total Fare: 800 Taka
	Passenger List:
	Walker Matt Wilson
	5----------
	Dear Henry! Welcome to Taxi

`t1.info();`
`System.out.println("9------------");`
`Taxi t2 = new Taxi();`
`t2.createTaxi(3);`
`System.out.println("10------------");`
`t2.info();`
`  }`
`}`
	`6------------`
	`Total Passenger: 4`
	`Maximum Capacity: 4`
	`Total Fare: 950 Taka`
	`Passenger List:`
	`Walker Matt Wilson Henry`
	`7------------`
	`Taxi is full`
	`8------------`
	`Total Passenger: 4`
	`Total Fare: 950 Taka`
	`Passenger List:`
	`Walker Matt Wilson Henry`
	`9------------`
	`A taxi is created with a capacity of 3`
	`10------------`
	`Total Passenger: 0`
	`Maximum Capacity: 3`
	`Total Fare: 0 Taka`
	`Passenger List:`

Task 9: **[Hometask]**

Write the design class with the required properties to produce the given output for the provided driver code.

Driver Code	Output

`public class CartTester{`
`  public static void main(String args[]){`
`  Cart c1 = new Cart();`
`  System.out.println("1------------");`
`  c1.info();`
`  System.out.println("2------------");`
`  Cart c2 = new Cart();`
`  System.out.println("3------------");`
`  c1.addItem("Coke", 12);`
`  c1.addItem("Biscuit", 50);`
`  System.out.println("4------------");`
`  c1.info();`
`  System.out.println("5------------");`
`  c1.addItem("Chocolate",15);`
`  System.out.println("6------------");`
`  c1.info();`
`  System.out.println("7------------");`
`  c1.addItem("Pen",5);`
`  System.out.println("8------------");`
`  c1.info();`
`  System.out.println("9------------");`
	`1------------`
	`Cart capacity: 3`
	`Total Item: 0`
	`Total Price: 0 Taka`
	`Item  List:`
	`2------------`
	`3------------`
	`Coke is added to cart.`
	`Biscuit is added to cart.`
	`4------------`
	`Cart capacity: 3`
	`Total Item: 2`
	`Total Price: 62 Taka`
	`Item  List:`
	`Coke Biscuit`
	`5------------`
	`Chocolate is added to cart.`
	`6------------`
	`Cart capacity: 3`
	`Total Item: 3`
	`Total Price: 77 Taka`

`    c2.addItem("Water",10);`
`    System.out.println("10------------");`
`    c2.info();`
`  }`
`}`
	`Item  List:`
	`Coke Biscuit Chocolate`
	`7------------`
	`Cart is full`
	`8------------`
	`Cart capacity: 3`
	`Total Item: 3`
	`Total Price: 77 Taka`
	`Item  List:`
	`Coke Biscuit Chocolate`
	`9------------`
	`Water is added to cart.`
	`10------------`
	`Cart capacity: 3`
	`Total Item: 1`
	`Total Price: 10 Taka`
	`Item  List:`
	`Water`

Task 10:

1	`public class Task11 {`
2	`   public int p = 3, y = 2, sum;`
3	`   public void methodA(){`
4	`       int x = 0, y = 0;`
5	`       y = y + this.y;`
6	`       x = sum + 2 + p;`
7	`       sum = x + y + this.methodB(p, y);`
8	`       System.out.println(x + " " + y+ " " + sum);`
9	`   }`
10	`   public int methodB(int p, int n){`
11	`       int x = 0;`
12	`       y = y + (++p);`
13	`       x = x + 2 + n;`
14	`       sum = sum + x + y;`
15	`       System.out.println(x + " " + y+ " " + sum);`

```
        return sum;
    }
}
```

Driver code:

```
public class Tester11 {
   public static void main(String [] args){
       Task11 t1 = new Task11 ();
       t1.methodA();
       t1.methodA();
   }
}
```

Outputs		
x	**y**	**Sum**

Task 11: **[Homework]**

```
public class A {
   public int temp = 4;
   public int sum, y, x;
   public void methodA(int m){
       int [] n = {2,5};
       int x = 0;
       y = m + this.methodB(x++,m)+(temp++);
       x = this.x + 2 + n[0];
       sum = sum + x + y;
       n[0] = sum + 2;
       System.out.println(n[0]+" " + x+ " " + sum);
   }
   public int methodB(int m, int n){
       int y = 4 + this.y + m;
       x = this.y + y + (++temp) - n;
       sum = x + y + this.sum;
```

17	`System.out.println(y+ " " + this.x + " " +sum);`
18	`return x;`
19	`  }`
20	`}`

	Outputs		

`public class Tester {`
`  public static void main(String [] args){`
`  A t1 = new A();`
`  t1.methodA(5);`
`  A t2 = new A();`
`  t2.methodB(12, 2);`
`  }`
`}`			

Method Overloading

Task 12:

Driver Class	Output

`public class Tester{`
`  public static void main(String [] args){`
`  Student s1 = new Student();`
`  Student s2 = new Student();`
`  Student s3 = new Student();`
`  Student s4 = new Student();`
`  s1.updateInfo("Ali");`
`  System.out.println("1----------");`
`  s1.details();`
`  System.out.println("2----------");`
`  s2.updateInfo("Joy", "MNS");`
`  s3.updateInfo("Akash", 3.99, "CS");`
`  s4.updateInfo("Rakin", "PHY", 3.89);`
`  System.out.println("3----------");`
`  s2.details();`
`  System.out.println("4----------");`
`  s3.details();`
`  }`
`}`
	`1----------`
	`Name: Ali`
	`Department: CSE`
	`CGPA: 0.0`
	`2----------`
	`3----------`
	`Name: Joy`
	`Department: MNS`
	`CGPA: 0.0`
	`4----------`
	`Name: Akash`
	`Department: CS`
	`CGPA: 3.99`

Task 13:

Driver Class	Output

`public class BakeryTester{`
`  public static void main (String args []){`
`  Bakery b1 = new Bakery();`
`  Bakery b2 = new Bakery();`
`  b1.setShop("Sweet Cravings");`
`  b2.setShop("Dreamy Delights", 7);`
`  b1.details();`
`  System.out.println("----------");`
`  b2.details();`
`  System.out.println("----------");`
`  b1.addItems("Cookies");`
`  b1.addItems("Donuts");`
`  b1.addItems("Muffins", 12);`
`  System.out.println("----------");`
`  b1.details();`
`  System.out.println("----------");`
`  b1.sellItems("Donuts",5);`
`  b1.sellItems("Cookies",7);`
`  System.out.println("----------");`
`  b1.sellItems("Cake",2);`
`  System.out.println("----------");`
`  b1.sellItems("Muffins",15);`
`  System.out.println("----------");`
`  b1.details();`
`  }`
`}`
	`Bakery Name: Sweet Cravings`
	`Product Capacity: 10`
	`Product Info:`
	`----------`
	`Bakery Name: Dreamy Delights`
	`Product Capacity: 7`
	`Product Info:`
	`----------`
	`20 Cookies are added.`
	`20 Donuts are added.`
	`12 Muffins are added.`
	`----------`
	`Bakery Name: Sweet Cravings`
	`Product Capacity: 10`
	`Product Info:`
	`1)Cookies: 20 pieces`
	`2)Donuts: 20 pieces`
	`3)Muffins: 12 pieces`
	`----------`
	`5 Donuts are sold.`
	`7 Cookies are sold.`
	`----------`
	`Item not found.`
	`----------`
	`Sorry! 15 Muffins are not available.`
	`----------`
	`Bakery Name: Sweet Cravings`
	`Product Capacity: 10`
	`Product Info:`
	`1)Cookies: 13 pieces`
	`2)Donuts: 15 pieces`
	`3)Muffins: 12 pieces`

Constructor & Constructor Overloading

Task 14:

Driver Class	Output

`public class BankAccountTester{`
`  public static void main(String args[]){`
`  BankAccount acc1 = new BankAccount();`
`  System.out.println("-----1-----");`
`  acc1.info();`
`  }`
`}`
	A new account has
	been created
	-----1-----
	Account No: 2324
	Type: Salary

Task 15:

`public class MobileTester{`
`  public static void main(String args[]){`
`  Mobile m1 = new Mobile();`
`  System.out.println("-----1-----");`
`  m1.brand = "Nokia";`
`  m1.simType = "Dual";`
`  System.out.println("-----2-----");`
`  m1.printDetails();`
`  }`
`}`
	A mobile is created
	-----1-----
	-----2-----
	Brand: Nokia
	Sim Type: Dual

Task 16:

`public class BankAccountTester{`
`  public static void main(String args[]){`
`  BankAccount acc1 = new BankAccount();`
`  System.out.println("-----1-----");`
`  acc1.printDetails();`
`  System.out.println("-----2-----");`
`  BankAccount acc2  = new BankAccount(5779,"Student");`
`  System.out.println("-----3-----");`
`  acc2.printDetails();`
`  }`
`}`
	-----1-----
	Account No: 0
	Type: Not Set
	-----2-----
	Bank account is created
	-----3-----
	Account No: 5779
	Type: Student

Task 17: [Homework]

Design the **Toy** class in such a way that it produces the following output

Driver Code	Expected Output

`public class ToyTester{`
`  public static void main(String[] args){`
`  Toy t1 = new Toy("Car", 230);`
`  System.out.println("1==========");`
`  t1.updatePrice(250);`
`  System.out.println("2==========");`
`  System.out.println(t1.name);`
`  t1.showPrice();`
`  System.out.println("3==========");`
`  Toy t2 = new Toy("Robot", 450);`
`  System.out.println("4==========");`
`  t2.updateName("Autobot");`
`  System.out.println("5==========");`
`  System.out.println(t2.name);`
`  t2.showPrice();`
`  }`
`}`
	`A new toy has been made!`
	`1==========`
	`2==========`
	`Car`
	`price: 250 Taka`
	`3==========`
	`A new toy has been made!`
	`4==========`
	`Changing old name: Robot`
	`new name: Autobot`
	`5==========`
	`Autobot`
	`price: 450 Taka`

Task 18:

`public class BakeryTester{`
`  public static void main (String args []){`
`  Bakery b1 = new Bakery("Sweet Cravings", 10);`
`  Bakery b2 = new Bakery("Dreamy Delights");`
`  b1.details();`
`  System.out.println("----------");`
`  b2.details();`
`  System.out.println("----------");`
`  b1.addItems("Cookies", 20);`
`  b1.addItems("Donuts", 15);`
`  b1.addItems("Muffins", 12);`
`  System.out.println("----------");`
`  b1.details();`
`  System.out.println("----------");`
`  b1.sellItems("Donuts",5);`
`  b1.sellItems("Cookies",7);`
`  System.out.println("----------");`
`  b1.sellItems("Cake",2);`
`  System.out.println("----------");`
	`Bakery Name: Sweet Cravings`
	`Product Capacity: 10`
	`Product Info:`
	`----------`
	`Bakery Name: Dreamy Delights`
	`Product Capacity: 7`
	`Product Info:`
	`----------`
	`20 Cookies are added.`
	`15 Donuts are added.`
	`12 Muffins are added.`
	`----------`
	`Bakery Name: Sweet Cravings`
	`Product Capacity: 10`
	`Product Info:`
	`1)Cookies: 20 pieces`
	`2)Donuts: 15 pieces`
	`3)Muffins: 12 pieces`
	`----------`

`b1.sellItems("Muffins",15);`
`System.out.println("----------");`
`b1.details();`
`}`
`}`
	5 Donuts are sold.
	7 Cookies are sold.
	---------- 	
	Item not found.
	---------- 	
	Sorry! 15 Muffins are not available.
	---------- 	
	Bakery Name: Sweet Cravings
	Product Capacity: 10
	Product Info:
	1)Cookies: 13 pieces
	2)Donuts: 10 pieces
	3)Muffins: 12 pieces

Task 19: [Homework]

Hint: CG >= 3 can add max 4 courses. Otherwise, max 3 courses

Driver Code	Expected Output

`public class StudentDriver {`
`public static void main(String[] args){`
`Student student1 = new Student(12345678);`
`System.out.println("1----------");`
`student1.addCourse("CSE110");`
`System.out.println("2----------");`
`student1.storeCG(2.5);`
`student1.addCourse("CSE110");`
`student1.addCourse("ENG101");`
`System.out.println("3----------");`
`student1.showAdvisee();`
`System.out.println("4----------");`
`student1.removeAllCourse();`
`System.out.println("5----------");`
`student1.showAdvisee();`
`System.out.println("6----------");`
`String[] courses = {"SOC101","CSE111","ENG102"};`
`student1.addCourse(courses);`
`System.out.println("7----------");`
`student1.showAdvisee();`
`System.out.println("8----------");`
`student1.addCourse("CSE230");`
	A student with ID 12345678 has been created.
	1----------
	Failed to add CSE110. Set CG first
	2----------
	Added CSE110
	Added ENG101
	3----------
	Student ID: 12345678, CGPA: 2.5
	Added courses are:
	CSE110 ENG101
	4----------
	All the courses are removed.
	5----------
	Student ID: 12345678, CGPA: 2.5
	No courses added.
	6----------
	Added SOC101
	Added CSE111
	Added ENG102
	7----------
	Student ID: 12345678, CGPA: 2.5

```
    student1.showAdvisee();
    System.out.println("9-----------------");
    Student student2 = new Student(975738383,3.0);
    System.out.println("10-----------------");
    student2.addCourse(new String []
{"CSE220","PHY112","MAT120","BUS101","CHN101"});
    System.out.println("11-----------------");
    student2.showAdvisee();
  }
}
```

```
Added courses are:
SOC101 CSE111 ENG102
8-----------------
Failed to add CSE230
CG is low. Can't add more than 3
courses.
Student ID: 12345678, CGPA: 2.5
Added courses are:
SOC101 CSE111 ENG102
9-----------------
A student with ID 975738383 and
cgpa 3.7 has been created.
10-----------------
Added CSE220
Added PHY112
Added MAT120
Added BUS101
Failed to add CHN101
Maximum 4 courses allowed.
11-----------------
Student ID: 975738383, CGPA: 3.0
Added courses are:
CSE220 PHY112 MAT120 BUS101
```

Task 20:

```
public class A{
    public int x = 4, y, z = 5;
    public double p = 0.0;
    public A(int x) {
        this.x = methodB(x++);
        p = x + this.x + z * 3.0;
        System.out.println(this.x +" " + x + " " + p) ;
    }
    public int methodB(int q, int n) {
        int arr[] = {3,4,5};
        arr[0] = arr[0] + this.x + n;
        arr[1] = q + arr[1];
        System.out.println(arr[0] +" " + arr[1] + " " + arr[2]) ;
        return arr[1] + arr[2];
    }
    public int methodB(int y) {
        if(y % 2 == 0) {
            return this.methodB(2, y);
        }
        else{
            return 4;
       }
    }
}
```

Driver Code	Output		

public class Tester11 {
public static void main(String [] args){
A a1;
a1 = new A(12);
a1.methodB(1,3);
new A(15).methodB(5,4); [Homework]
}
}			

Reference Passing & Multiclass Problems

Task 21: [Classwork]

Driver Code	Output

`import java.util.Arrays;`
`public class Tester{`
`  public static void main(String [] args){`
`  Department cse = new Department("CSE", 1200);`
`  Department cs = new Department("CS", 700);`
`  System.out.println("1--------------------");`
`  cse.info();`
`  System.out.println("2--------------------");`
`  cs.info();`
`  System.out.println("3--------------------");`
`  cse.mergeStudents(cs);`
`  System.out.println("4--------------------");`
`  cse.info();`
`  System.out.println("5--------------------");`
`  int [] newStudents = {12, 13, 16};`
`  cse.addStudents(newStudents);`
`  System.out.println("6--------------------");`
`  cse.info();`
`  System.out.println("7--------------------");`
`  System.out.println(Arrays.toString(newStudents));`
`  }`
`}`
`//cse.addStudents(new int [] {12,13,14});`
	`1--------------------`
	`Name: CSE`
	`Total Students: 1200`
	`2--------------------`
	`Name: CS`
	`Total Students: 700`
	`3--------------------`
	`Students of CS are merged`
	`with students of CSE`
	`4--------------------`
	`Name: CSE`
	`Total Students: 1900`
	`5--------------------`
	`6--------------------`
	`Name: CSE`
	`Total Students: 1941`
	`7--------------------`
	`[0, 0, 0]`

Task 22: [Homework]

Driver Code	Output

`public class TaxiTester{`
`  public static void main(String args []){`
`  Taxi t1 = new Taxi("MX8190", "Gulshan");`
`  Taxi t2 = new Taxi("MX6718", "Badda");`
`  Taxi t3 = new Taxi("MX2789", "Gulshan");`
`  System.out.println("1------------");`
`  t1.compareTaxi(t2);`
`  System.out.println("2------------");`
`  t1.compareTaxi(t3);`
`  }`
`}`
	`1------------`
	`Routes are not the same!`
	`2------------`
	`Both routes are the same!`

Task 23: [Classwork]

Driver Code	Output

`public class Tester{`
`  public static void main(String args[]){`
`  Cart c1 = new Cart();`
`  Cart c2 = new Cart();`
`  Cart c3 = new Cart();`
`  Cart cartHub = new Cart();`
`  c1.storeProduct("Pen", 10);`
`  c2.storeProduct("Pencil", 15, "Book", 100);`
`  c2.storeProduct("Eraser", 5);`
`  c3.storeProduct("Sharpener", 8);`
`  System.out.println("1------------");`
`  c1.info();`
`  System.out.println("2------------");`
`  c2.info();`
`  System.out.println("3------------");`
`  c1.transferItem(c2, "Pen");`
`  System.out.println("4------------");`
`  c1.info();`
`  System.out.println("5-----------");`
`  c2.info();`
`  System.out.println("6-----------");`
`  cartHub.addCart(c1,c2,c3);`
`  System.out.println("7-----------");`
`  cartHub.allCartInfo();`
`  }`
`}`
	`1------------`
	`Cart Items:`
	`Pen - 10`
	`2------------`
	`Cart Items:`
	`Pencil - 15`
	`Book - 100`
	`Eraser - 5`
	`3------------`
	`Item transferred successfully.`
	`4------------`
	`Cart Items:`
	`Cart is empty.`
	`5-----------`
	`Cart Items:`
	`Pencil - 15`
	`Book - 100`
	`Eraser - 5`
	`Pen - 10`
	`6-----------`
	`7-----------`
	`Cart Items:`
	`Cart is empty.`
	`Cart Items:`
	`Pencil - 15`
	`Book - 100`
	`Eraser - 5`
	`Pen - 10`
	`Cart Items:`
	`Sharpener - 8`

Task 24: [Homework] [You can skip it for Quiz 2]

Driver Code	Output

`public class TaxiTester{`
`  public static void main(String args []){`
`  Taxi t1 = new Taxi("MX8190", "Gulshan");`
`  Passenger p1 = new Passenger("Ali", 20);`
`  Passenger p2 = new Passenger("Mifta", 45);`
`  Passenger p3 = new Passenger("Marzan", 32);`
`  System.out.println("1------------");`
`  t1.showTaxi();`
`  System.out.println("2------------");`
`  p1.info();`
`  p2.info();`
`  System.out.println("3------------");`
`  t1.addPassenger(p1);`
`  t1.addPassenger(p2);`
`  t1.addPassenger(p3);`
`  System.out.println("4------------");`
`  t1.showTaxi();`
`  }`
`}`
	`1------------`
	`Taxi No: MX8190`
	`Route: Gulshan`
	`Passenger List:`
	`2------------`
	`Name: Ali`
	`Age: 20`
	`Name: Mifta`
	`Age: 45`
	`3------------`
	`4------------`
	`Taxi No: MX8190`
	`Route: Gulshan`
	`Passenger List:`
	`Ali 20`
	`Mifta 45`
	`Marzan 32`

Task 25: [Homework]

Design the **Shop** class with the necessary properties so that the following output is produced.

Driver Code	Output

`public class ShopTester{`
`  public static void main(String[] args){`
`  Shop s1 = new Shop("Agora", 4);`
`  Shop s2 = new Shop(3);`
`  System.out.println("1==========");`
`  s1.info();`
`  System.out.println("2==========");`
`  s2.info();`
`  System.out.println("3==========");`
`  s1.addItem("KitKat", 25);`
`  s1.addItem("Hershey's", 100);`
`  s1.addItem("Dairy Milk", 150);`
`  s1.addItem("Feastables", 500);`
	`Agora shop created!`
	`Mega shop created!`
	`1==========`
	`Shop Name: Agora`
	`Item Details:`
	`No items in shop`
	`2==========`
	`Shop Name: Mega`
	`Item Details:`
	`No items in shop`
	`3==========`
	`Prime, could not be added`
	`4==========`

`s1.addItem("Prime", 1200);`
`System.out.println("4==========");`
`s1.info();`
`System.out.println("5==========");`
`s1.purchase("KitKat");`
`s1.purchase("Snickers");`
`System.out.println("6==========");`
`s2.info();`
`System.out.println("7==========");`
`s2.addItem(s1, "KitKat");`
`s2.addItem("Toblerone", 80);`
`s2.addItem(s1, "Mars");`
`System.out.println("8==========");`
`s2.info();`
`}`
`}`
	`Shop Name: Agora`
	`Item Details:`
	`4 / 4`
	`KitKat - 25 Tk`
	`Hershey's - 100 Tk`
	`Dairy Milk - 150 Tk`
	`Feastables - 500 Tk`
	`5==========`
	`Purchase Complete!`
	`Snickers, is not available in this shop`
	`6==========`
	`Shop Name: Mega`
	`Item Details:`
	`No items in shop`
	`7==========`
	`Mars, not found in Agora!`
	`8==========`
	`Shop Name: Mega`
	`Item Details:`
	`2 / 3`
	`KitKat - 25 Tk`
	`Toblerone - 80 Tk`

Task 26: [Homework]

Design the **Player** class to obtain the following output from the driver code. You can assume, each player can defeat at most 5 villains.

Driver Code	Expected Output

`public class PlayerTester {`
`public static void main(String[] args) {`
`Player ben = new Player("Ben", 10);`
`System.out.println("======1======");`
`ben.viewInfo();`
`System.out.println("======2======");`
`ben.defeatVillain("Vilgax", 100);`
`System.out.println("======3======");`
`ben.defeatVillain("Yamcha", 10);`
`System.out.println("======4======");`
`ben.viewInfo();`
`System.out.println("======5======");`
`ben.defeatVillain('8', "Vilgax", 100);`
`System.out.println("======6======");`
`ben.viewInfo();`
`System.out.println("======7======");`
`Player kevin = new Player("Kevin");`
	`Ben joined the game`
	`HP: 10`
	`======1======`
	`Player Name: Ben`
	`Current HP: 10`
	`======2======`
	`failed to defeat Vilgax`
	`======3======`
	`defeated Yamcha`
	`======4======`
	`Player Name: Ben`
	`Current HP: 20`
	`Defeated:`
	`Yamcha,`
	`======5======`
	`HP with 8x boost: 160`
	`defeated Vilgax`

`    System.out.println("======8======");`
`    kevin.viewInfo();`
`    System.out.println("======9======");`
`    ben.defeatVillain(kevin);`
`    System.out.println("======10======");`
`    ben.viewInfo();`
`    System.out.println("======11======");`
`    Player goku = new Player("Goku", 9000);`
`    System.out.println("======12======");`
`    ben.defeatVillain(goku);`
`  }`
`}`
	`======6======`
	`Player Name: Ben`
	`Current HP: 260`
	`Defeated:`
	`Yamcha, Vilgax,`
	`======7======`
	`Kevin joined the game`
	`HP: 100`
	`======8======`
	`Player Name: Kevin`
	`Current HP: 100`
	`======9======`
	`defeated Kevin`
	`======10======`
	`Player Name: Ben`
	`Current HP: 360`
	`Defeated:`
	`Yamcha, Vilgax, Kevin,`
	`======11======`
	`Goku joined the game`
	`HP: 9000`
	`======12======`
	`failed to defeat Goku`

Task 27: [Classwork]

Write the **Teacher** and **Course** classes so that the TestTeacher class produces the outputs given.
Hint: A teacher can add a maximum of 3 courses.

Driver Code	Output

`public class TestTeacher{`
`  public static void main(String [] args){`
`  Teacher t1 = new Teacher("Matin Saad Abdullah","MSA");`
`  Teacher t2 = new Teacher("Mumit Khan","MMK");`
`  Teacher t3 = new Teacher("Sadia Hamid Kazi","SKZ");`
`  Course c1 = new Course("CSE 110");`
`  Course c2 = new Course("CSE 111");`
`  Course c3 = new Course("CSE 220");`
`  Course c4 = new Course("CSE 221");`
`  Course c5 = new Course("CSE 230");`
`  Course c6 = new Course("CSE 310");`
`  Course c7 = new Course("CSE 320");`
`  Course c8 = new Course("CSE 340");`
`  t1.addCourse(c1);`
`  t1.addCourse(c2);`
`  t2.addCourse(c3);`
`  t2.addCourse(c4);`
`  t2.addCourse(c5);`
`  t3.addCourse(c6);`
`  t3.addCourse(c7);`
`  t3.addCourse(c8);`
`  System.out.println("1=======================");`
`  t1.printDetail();`
`  System.out.println("2=======================");`
`  t2.printDetail();`
`  System.out.println("3=======================");`
`  t3.printDetail();`
`  }`
`}`
	`A new teacher has been created`
	`A new teacher has been created`
	`A new teacher has been created`
	`1=======================`
	`Name: Matin Saad Abdullah`
	`Initial: MSA`
	`List of courses:`
	`CSE 110`
	`CSE 111`
	`2=======================`
	`Name: Mumit Khan`
	`Initial: MMK`
	`List of courses:`
	`CSE 220`
	`CSE 221`
	`CSE 230`
	`3=======================`
	`Name: Sadia Hamid Kazi`
	`Initial: SKZ`
	`List of courses:`
	`CSE 310`
	`CSE 320`
	`CSE 340`

Task 28: [Classwork]

Design the required class/es so that the following output is generated. Read the following description:

1. You may assume that to board a bus, a student must have the bus pass, and his/her location must match the route of the bus.
2. Additionally, the default maximum capacity of the bus is 2.
3. The location attribute of the Student class will be **private**

Driver Code	Output

`public class BracuStudentTester {`
` public static void main(String[] args) {`
`  BracuStudent st1 = new BracuStudent("Afif", "Mirpur");`
`  System.out.println("1==========");`
`  BracuStudent st2 = new BracuStudent("Shanto",`
`"Motijheel");`
`  BracuStudent st3 = new BracuStudent("Taskin", "Mirpur");`
`  st1.showDetails();`
`  st2.showDetails();`
`  System.out.println("2==========");`
`  st3.showDetails();`
`  System.out.println("3==========");`
`  BracuBus bus1 = new BracuBus("Mirpur");`
`  BracuBus bus2 = new BracuBus("Azimpur", 5);`
`  bus1.showDetails();`
`  bus2.showDetails();`
`  System.out.println("4==========");`
`  st2.collectPass();`
`  st3.collectPass();`
`  System.out.println("5==========");`
`  st2.showDetails();`
`  st3.showDetails();`
`  System.out.println("6==========");`
`  bus1.board();`
`  System.out.println("7==========");`
`  bus1.board(st1, st2);`
`  System.out.println("8==========");`
`  st1.collectPass();`
`  st2.setLocation("Mirpur");`
`  st1.showDetails();`
`  st2.showDetails();`
`  System.out.println("9==========");`
`  bus1.board(st1);`
	`1==========`
	`Student Name: Afif`
	`Lives in Mirpur`
	`Have Bus Pass? false`
	`Student Name: Shanto`
	`Lives in Motijheel`
	`Have Bus Pass? false`
	`2==========`
	`Student Name: Taskin`
	`Lives in Mirpur`
	`Have Bus Pass? false`
	`3==========`
	`Bus Route: Mirpur`
	`Passenger Count: 0 (Max: 2)`
	`Passengers on Board:`
	`Bus Route: Azimpur`
	`Passenger Count: 0 (Max: 5)`
	`Passengers on Board:`
	`4==========`
	`5==========`
	`Student Name: Shanto`
	`Lives in Motijheel`
	`Have Bus Pass? true`
	`Student Name: Taskin`
	`Lives in Mirpur`
	`Have Bus Pass? true`
	`6==========`
	`No passengers`
	`7==========`
	`You don't have a bus pass!`
	`You got on the wrong bus!`
	`8==========`
	`Student Name: Afif`
	`Lives in Mirpur`
	`Have Bus Pass? true`
	`Student Name: Shanto`
	`Lives in Mirpur`
	`Have Bus Pass? true`

`  bus1.board(st2, st3);`
`  System.out.println("10==============");`
`  bus1.showDetails();`
`  }`
`}`
	`9==============`
	`Afif boarded the bus.`
	`Shanto boarded the bus.`
	`Bus is full!`
	`10==============`
	`Bus Route: Mirpur`
	`Passenger Count: 2 (Max: 2)`
	`Passengers on Board:`
	`Afif Shanto`

Task 26: [Classwork]

```
pubic class QuizA {
  public int x, y, sum = 1;
  public QuizA(int x, int y) {
    this.x = y;
    this.y = x;
  }
  public void methodA() {
    int x = 3;
    y = this.y + x;
    QuizA exam = new QuizA(5, 11);
    exam.sum = x;
    exam.y = this.y;
    x = this.x + x + exam.sum;
    this.y = this.sum + methodB(exam.sum, exam);
```

```
    System.out.println(exam.x + " " + this.y + " " + sum);
  }
  public int methodB(int x1, QuizA x2) {
    int y = 0;
    y = this.y + x2.sum;
    x2.sum = x1 + x2.x;
    sum = sum + x + y;
    System.out.println(this.x + " " + this.y + " " + sum);
    return x2.sum;
  }
  public static void main(String []args){
    QuizA q1 = new QuizA(3,4);
    q1.methodA();
  }
}
```

Task 24: [Classwork] [You can skip it for Quiz 2]

```
class Exam{
  public int y;
  public int sum;
  public int x = 100;
  public void methodA() {
    System.out.println(y + " " + 0 + " " + 0);
  }
}

class QuizA {
  public int x, y;
  public int sum = 1;

  public QuizA(int x, int y) {
    this.x = y;
    this.y = x;
  }
  public void methodA() {
    int x = 3;
    y = this.y + x;
    Exam ex = new Exam();
    ex.sum = x;
    ex.y = this.y;
    x = this.x + x + ex.sum;
    this.y = this.sum + methodB(3, ex) + this.x
    System.out.println(x + " " + this.y + " " + sum);
    sum = x % 2 + this.x;
    y = x + y + ex.sum;
    System.out.println(x + " " + y + " " + sum);
  }
  public int methodB(int x1, Exam x2) {
    int y = 0;
    y = this.y + x2.sum;
    x2.sum = x1 + x2.x;
    sum = sum + x + y;
    System.out.println(this.x + " " + this.y + " " + sum);
    return x2.sum;
  }
```

```
}

public class T{
  public static void main(String []args){
    QuizA q1 = new QuizA(3,4);
    q1.methodA();
  }
}
```